

Toward Deployment

The cluster: three projects, one borrowed word, and an adjective that is never true on its own

Sim2real is a borrowed word and I borrowed it on purpose. In robotics the gap between where a policy is trained and where it runs is a gap in physics: friction, latency, sensor noise, the mass of a gripper. The shape of my problem is the same. The content is not. This file says what the word carries once it arrives in a network, and it fixes the half-dozen terms the three projects under this heading all use, because each of them took its vocabulary from a different place and none of us negotiated.

The rest stays with the project it belongs to. Metrion is in `td_realistic_cyber_environment`, FOE-Dreamer on Daedalus is in `td_training_in_realistic_environments`, and the transfer work is in `td_sim2sim_before_sim2real`. The word *environment* itself is not defined here at all. It does six separate jobs across my files and the census of those six is `dict_cyber_env`. Everything below assumes it.

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1 The borrowed word

The cluster page states the problem in four sentences and every term in this file is an answer to one of them.

“Sim2real refers to the long standing challenge encountered by embodied agents, between where a policy is trained and where it would run.”

website, Toward Deployment, Sim2Real Challenges in Cybersecurity

“Dynamics and sensing differ between the training world and the real one.”

website, Toward Deployment

“A policy drifts as the deployment ages away from the data it was built on.”

website, Toward Deployment

“And any safety we claim has to hold while the model is wrong, because out of distribution it will be.”

website, Toward Deployment

Three sentences, three failures, and they are not the same failure. The first is a gap in space: the training world is not the deployment world. The second is a gap in time: the deployment world is not the world the policy was fitted to, because the network kept moving after the snapshot was taken. The third is not a gap at all. It is a requirement on behaviour in the region where both of the first two have already gone wrong. Each of the three open questions on the Next page answers exactly one of them, in that order.

2 What the cyber setting does to it

Four things change when the word crosses over, and each one costs a project.

There is no physics to converge to. A robot’s real world is governed by laws that hold whether or not anyone wrote them down, so a simulator can be wrong about them and can be made less wrong. A network is an artifact. It is whatever some organization built, configured, patched unevenly and then partly forgot, and two enterprises differ from each other more than two laboratory floors do. There is no single target to approach, which is why the question *is this environment realistic* has no answer and the question *realistic for what* does. That reframing is the whole of Metrion.

The real world has an occupant who wants to be missed. Friction does not adapt. An adversary does, and the parts of the environment that make adaptation possible, benign activity to blend into and telemetry to evade, are the parts current environments most often leave out. So the sim2real gap here is not just a modelling error, it is a modelling error that one of the two players is actively exploiting.

There is no cheap real to transfer into. You do not get to try the policy on the production network and see what happens. That is the sentence the whole cluster sits on, and it is stated flatly on the third project’s page: letting a learning agent explore production infrastructure by trial and error is not an option. Everything else follows. Either you make training in a real enough environment affordable, which is the second project, or you learn to measure the gap without an

operational network to measure against, which is the third.

The target is only partially known, and not by anyone. In robotics you can at least instrument the real system. Here the defender’s own inventory is incomplete, the segmentation is not what the diagram says, and the organization will not hand over its topology because it cannot, for privacy reasons and sometimes because nobody has it. That is what makes sim2real strictly harder than sim2sim rather than a different problem: supply the missing ground truth and it collapses back to sim2sim, which is still hard.

3 The three projects

Metrion

The taxonomy of realism. It refuses the adjective as posed and replaces it with a pair: an environment and an evaluation objective, scored against the properties that objective needs. Eleven dimensions in five groups, 115 scoring elements, a fit score, and one hard rule that overrides the score. Poster at ACM CCS 2026 with Maria Rigaki and Carlos A. Catania, *Realistic Enough for What? Metrion: A Multidimensional Framework for Evaluating Cyber Environments*. It contributes the whole vocabulary of dimensions and requirements, and the position that suitability belongs to a pair. Full entry in `td_realistic_cyber_environment`.

FOE-Dreamer, on Daedalus

The other route. Rather than transfer a policy out of a simulator, train the defender inside an emulated operational network and pay for it in sample efficiency. Daedalus is the eight-host OpenStack network it trains in, with real services, a gRPC command-and-control server that carries each defender action to the host it names, a scripted red agent running a genuine web exploit, and generated background user activity. Two backends behind one interface, a fast simulator for the millions of steps and the live deployment for evaluation. Preprint, *FOE-Dreamer: Deployment-Efficient Learning of Cyber Defense Policies in Operational Networks*, with Christopher Kiekintveld, under review at ACSAC. It contributes *backend*, *fidelity gap*, and the scoped reading of *realistic*. Full entry in `td_training_in_realistic_environments`.

Sim2Sim before Sim2Real

The reduction. If the sim2real gap cannot be measured without an operational network, transform it into a sim2sim gap, which loses none of the difficulty and can be run today. Two mappings built without touching either simulator, a shared action vocabulary indexed by kill-chain stage and a fixed-size state projection, with a light encoder trained by adversarial domain confusion to close what is left. *Crossing the Cyber Divide: Sim-to-Sim and Sim-to-Real Transfer for RL Agents*, RAISE workshop at ESORICS 2026, with Sabrina Saika and Aritran Piplai. It contributes *action translation*, *state alignment* and the offensive sense of *transfer*. Full entry in `td_sim2sim_before_sim2real`.

The three are not three attempts at one thing. Metrion says what a claim needs, FOE-Dreamer buys one class of claim outright by paying the live cost, and Sim2Sim measures the gap for everyone who cannot pay it. Read as a vocabulary statement: Metrion owns the requirement words, FOE-Dreamer owns the cost words, Sim2Sim owns the alignment words. Both of the disagreements recorded below are words that crossed one of those boundaries and kept their old sense on arrival.

4 The shared terms

Sixteen terms, each used by at least two of the three projects. Four fields apiece. Where two of my files use one term two ways the disagreement sits inside the entry, because that is the entry worth having.

4.1 The two reductions

sim2real

“the long standing challenge encountered by embodied agents, between where a policy is trained and where it would run”

website, Toward Deployment

Definition: The problem of running a policy in a deployment whose dynamics, sensing and composition differ from the training environment, where the deployment cannot be enumerated in advance and cannot be used for exploratory training.

Distinguishes it from: Ordinary distributional shift, which is a statement about two distributions. Sim2real names the practical situation that produces the shift, and adds two constraints a shift statement does not carry: the target is only partially observable, and errors there are expensive.

Operational test: Can you enumerate the target environment? If yes, this is sim2sim. If no, and you also cannot train in it, this is sim2real.

In my project: The heading the cluster is named for. Metrion attacks it by refusing to compare an environment against *the* real world at all. FOE-Dreamer attacks it by deleting the simulator stage. Sim2Sim attacks it by converting it into a problem that can be measured.

sim2sim

“the sim2real challenge can be transformed to a sim2sim without loss of its complexity”

website, Toward Deployment, Where We Are at Year-1

Definition: Transfer between two fully specified environments with non-coinciding state spaces, action vocabularies and observability assumptions. Sim2real with the missing ground truth supplied.

Distinguishes it from: Domain adaptation in the usual sense, where the two domains share a task specification and differ in appearance. Here the action sets themselves differ, so there is no shared vocabulary to adapt within until one is constructed.

Operational test: Can you write down both environments in full? Then the remaining difficulty is finding the abstraction that relates them, which is by itself computationally hard.

In my project: The claim that makes the third project possible. It is a claim about complexity, not about convenience: the reduction is asserted to lose none of the difficulty, and that assertion is what the transfer results have to support.

The reduction runs one way and the word does not

The README says the sim2real challenge *can be transformed to* a sim2sim. The Sim2Sim page says sim-to-real *is the same problem as* sim-to-sim with one extra difficulty.

Those are different claims. A transformation into an easier problem is a research strategy and gives up nothing only if the reduction is tight. An identity plus an extra difficulty is a statement that the easier problem is a strict sub-part. Both sentences are mine, they were written for different pages, and the second is the one the experiments actually test. Anything I publish should say which, since a reviewer who reads the first will ask what was lost in the transformation and the second sentence already answers that it is the partial knowledge of the target.

4.2 The adjective

realistic

“An environment is not realistic or unrealistic; it is realistic in some respects and not others, and which respects it got right is what decides whether a result transfers.”

website, Training in “Realistic” Environments, what “realistic” is doing in that sentence

Definition: A two-place predicate, never a one-place one. An environment is realistic *in a named respect*, and useful only *for a named claim*. The quotation marks in two of the three page titles are not decoration.

Distinguishes it from: Fidelity, which in my files is a property of one named layer, and from validity, which is a property of an evaluation rather than of an environment. An environment can be high fidelity on every layer a claim does not depend on.

Operational test: Ask what would make this environment more realistic. If three people answer noise and a moving opponent, Active Directory and event forwarding, and task pressure with a cover story the participant believes, then the word has three senses in the room and the conversation will not converge.

In my project: Metrion turns the two places into a scoring procedure. FOE-Dreamer states them by hand: the substrate, the services, the exploit paths, the five-second polling and the consequences of defender actions are real, while the user and attacker populations and the CVE catalogue are emulated. The two positions are consistent. One is the general form and one is a filled-in instance.

realism, fidelity

“Verification and validation of a model of a natural system are impossible in principle, because such systems are never closed and model results are never unique.”

website, When We Say “A Realistic Cyber Environment”, why “is it realistic?” is the wrong question

Definition: In the realism evaluation, *fidelity* is the score of a single named layer, as in service fidelity, OS fidelity, telemetry fidelity. *Realism* is what the whole gets, and what each of the eleven dimensions is called when it is not a fidelity.

Distinguishes it from: The blog usage, which reverses the two. On the Cyber Environments

pages fidelity is the aggregate, as in fidelity and tractability pull against each other. There is no third sense, only two dialects with the labels swapped.

Operational test: Count the layers the sentence is about. One layer, fidelity. Eleven, realism. If the sentence names none, it is an aggregate claim and should say so.

In my project: Metrion uses the split deliberately, three fidelities and eight realisms across eleven dimensions. The cost is that a sentence like *aims at the highest fidelity short of deploying the real thing* reads, to anyone coming from the evaluation, as a claim about eleven dimensions at once, and it is not one.

Two dialects, reversed labels

The blog corpus says fidelity five times and never says realism. The realism evaluation says both and splits them, with thirty-five of its thirty-eight fidelity tokens sitting inside the three dimension names.

So both corpora carry the same distinction between a layer and the whole, and they attach the labels the opposite way round. This is recorded in full in `dict_cyber_env`. It matters here because all three project pages in this cluster inherit the blog dialect while the framework that scores them uses the other, and the two sit one click apart on the site.

4.3 Transfer, and the gap it leaves

transfer

“can a policy trained to competence in one be executed in the other without retraining?”

website, Sim2Sim before Sim2Real, the problem

Definition: Executing a policy in an environment other than the one it was fitted in, with no gradient steps taken in the destination.

Distinguishes it from: Fine-tuning, which is retraining with a warm start and answers a different question, and from generalization within one environment, where the interface does not change.

Operational test: Were any parameters updated after the destination was first seen? If yes it is not transfer in this cluster’s sense.

In my project: The word carries a different object in two of the three projects and the difference is not cosmetic. See the box.

One word, two objects, two directions

On the Sim2Sim page the thing transferred is an *offensive* policy and it moves *across simulators*: CyberWheel as source, NetSecGame and CyberBattleSim as targets at different distances, NASim’s emulation mode as a proxy for deployment. The obstacle is representational, two environments that describe the world differently.

On the Training page and throughout Daedalus the thing transferred is a *defensive* policy and it moves *between two backends of one environment*, from the fast simulator to the live emulated deployment, with the action and observation spaces held identical by construction. The obstacle is not representational at all. It is fidelity: what the simulator abstracted away.

Same word, opposite sides of the engagement, and the difficulty is located in different places. The first needs an alignment. The second needs no alignment and gets a measurement instead. Any page that puts both under one heading has to say which is meant in the first clause.

fidelity gap

“the actions that worked in the simulator and failed against the real service stack are the fidelity gap made concrete”

artifact, Daedalus, use

Definition: The difference in outcome between the same action taken in the cheap backend and in the honest one. Not an error term to be minimized but a quantity to be read off.

Distinguishes it from: Model error in the world-model sense, which is a property of something fitted from data. The fidelity gap is a property of two hand-built things and is fully attributable: you can name the action, the host, and the abstraction that caused it.

Operational test: Run the same policy on both backends and diff the outcomes per action. What differs is the gap, and it is a list rather than a number.

In my project: The reason the two-backend design is a result and not just an engineering convenience. Evaluation on Daedalus reports where the simulation lied, not only whether the defence held. NASimEmu and PenGym report the same phenomenon from the offensive side and are the two citations Metrion opens with.

action translation

“A shared, deterministic action vocabulary in which each action advances a chosen host along the kill chain.”

website, Sim2Sim before Sim2Real, the approach

Definition: A fixed intermediate action set, indexed by kill-chain stage and host, with per-environment wrapper logic resolving each choice into whatever the destination simulator natively expects. Nothing is learned.

Distinguishes it from: A learned action embedding, which would need trajectories from the destination. This needs none, and it separates the policy’s intent, which host to move on next, from the mechanics of saying so.

Operational test: Could you write the wrapper for a new simulator by reading its documentation, without collecting a single episode? If not, it is not this.

In my project: One of the two mappings the third project is built from, and the half that handles structural mismatch. Full entry in `td_sim2sim_before_sim2real`.

state alignment

“Each raw observation is projected into a fixed-size vector organized by kill-chain stage per host”

website, Sim2Sim before Sim2Real, the approach

Definition: A deterministic per-environment projection of raw observations into one fixed schema, phase and reachability and attacker presence and target designation per host, plus a

small block of episode-level context. The learned encoder that follows it is a separate thing and closes the distributional gap only.

Distinguishes it from: Action translation, which fixes the output side. Also from the encoder: the projection is written by hand and the encoder is trained with adversarial domain confusion, reconstruction, and a mean-alignment term.

Operational test: Does the same feature index mean the same thing in both environments? That is the projection’s job. Does it carry comparable numbers? That is the encoder’s.

In my project: The other mapping. The detail worth carrying is the exception: the flags identifying the goal bypass the encoder entirely, because routing them through adversarial alignment suppresses the goal signal and the agent stops behaving goal-directedly.

4.4 Cost

backend

“Two backends sit behind one action and observation interface: a fast simulator for the millions of steps reinforcement learning requires, and the live emulated deployment for evaluation.”

website, Training in “Realistic” Environments, the environment it trains in

Definition: One of two implementations of a single action and observation interface, differing in what a step costs and in what a step actually does.

Distinguishes it from: Two environments, which is what the third project faces. Two backends share the interface by construction, so a policy moves between them with no alignment at all, and any difference in outcome is attributable to fidelity rather than to representation.

Operational test: Does the agent’s code change between the two? If yes they are environments, not backends.

In my project: The design that makes the fidelity gap measurable, and the reason *transfer* means something narrower here than on the Sim2Sim page.

sample efficiency

“Every step there costs seconds of wall-clock and a real state change on a real host, so what limits the result is sample efficiency, not asymptotic performance.”

website, Training in “Realistic” Environments, the experiment

Definition: Return achieved per unit of environment interaction, where the unit is priced in wall-clock seconds and in irreversible state changes on real machines rather than in simulator steps.

Distinguishes it from: Compute efficiency, which is about the GPU, and from asymptotic performance, which assumes the interaction is free. Here the two budgets are not interchangeable: three days on one GPU is the headline because the GPU was never the constraint.

Operational test: Halve the step budget. If the result survives, the method was not sample-limited and this is not the axis to report.

In my project: The claim the second project actually makes. The loss numbers are the evidence,

the budget is the finding, and the finding is that training in an emulated network is an option rather than a thought experiment.

operational

Definition: Stipulated narrowly, and only about how the environment is realized: services execute rather than being modelled, actions produce real state changes on real operating systems, and time passes on a wall clock. It is not a claim that the network belongs to anyone or carries production traffic.

Distinguishes it from: Production, which nobody in this cluster trains in and nobody claims to. Also from realistic, which is the two-place adjective above and is scored per respect.

Operational test: If you deleted the scenario, would there still be machines running services? That is operational. Whether anyone depends on them is a separate question and the answer here is no.

In my project: The word in the FOE-Dreamer title and in the five-year vista heading. Used loosely it would promise the thing the whole cluster exists because we cannot have.

4.5 After deployment

drift

“A policy is built on a snapshot; the network it defends keeps moving.”

website, Next, drift after deployment

Definition: The widening of the gap between the data a deployed policy was fitted to and the network it now acts in, driven by the network changing rather than by the policy moving.

Distinguishes it from: The sim2real gap, which is present at time zero and is a property of two environments. Drift is a property of one environment across time, and it accrues after the transfer question has already been answered.

Operational test: Freeze the policy and wait. Any degradation is drift. Anything present on day one is the transfer gap.

In my project: The second of the three open questions, and the only one stated as a detection problem: watching for the distribution shift that matters to *this* policy rather than shift in general, and deciding when to retrain, roll back, or hand control to a human.

out of distribution

“Out of distribution the model is wrong, and safety has to mean something anyway.”

website, Next, behavior while the model is wrong

Definition: The region where the model’s error is not small and cannot be assumed small. Treated here as the normal operating condition rather than as an edge case.

Distinguishes it from: Uncertainty that a well-calibrated model reports, which is a quantity inside the model. This is the region where the calibration itself is untrustworthy.

Operational test: Would you accept the model’s own confidence as evidence here? If not, you

are in it.

In my project: The third open question, and the one that ties this thread back to the faulty-model question in Cyber World Modeling. A deployed defender and a defender with a wrong world model are facing the same danger from two sides, so the graceful-degradation requirement is shared between two clusters and should not be written twice.

zero-shot

“Can offensive cyber policies transfer zero-shot across environments, with no retraining?”

website, Sim2Sim before Sim2Real, questions being asked

Definition: Executed in the destination with no parameter updates and no reward signal from it. Observations from a random policy are permitted, since encoder pretraining needs only those.

Distinguishes it from: Few-shot, which permits a small number of episodes in the destination, and from zero-day, which is a property of a vulnerability and not of a learning regime. The two words share a prefix and nothing else, and my Headspace page asks whether defence against zero-days is a zero-shot or a few-shot challenge, which is exactly the confusion worth keeping separate.

Operational test: Count the gradient steps taken after the destination was first seen. Zero, or the claim is something else.

In my project: The first of the three research questions on the Sim2Sim page, and the strictest one. The second question, which parts of a policy actually transfer, is the one that decides whether a positive answer to the first is informative.

deployment-readiness

“Transfer is only evidence of deployment-readiness if it survives being actively doubted.”

website, Next, winning for the right reasons

Definition: The property that a policy’s score in a destination environment was produced by the task rather than by the destination’s regularities. Defined by what it survives, not by a threshold.

Distinguishes it from: Transfer performance, which is a number, and from suitability in the Metrion sense, which is a property of an environment and an objective rather than of a policy.

Operational test: Three probes, and they are the open question rather than a settled protocol: causal probes that perturb the parts of the environment a genuine policy should depend on and a benchmark-reader should not, held-out kinds of realism the policy never trained against, and counterfactual episodes built to fail a policy that has memorised surface structure.

In my project: The first of the three open questions, and the one that connects this cluster to gain attribution in Metrion. A policy that has learned the benchmark and an environment that hands the policy its answer are the same failure seen from the two ends.

5 What stays in the project files

`td_realistic_cyber_environment`: the eleven dimensions and their five groups, scoring element, requirement profile, critical and useful and not needed, full and partial and absent and unknown, fit score, the hard constraint, suitable and partially suitable and incomplete, gain attribution, partial confirmation, the use-case catalog and the group archetypes, the two-layer split between environment realism and scenario validity, the opponent-model scale, and the implementation-against-usage ontology that separates simulator from emulator from cyber range from benchmark from testbed.

`td_training_in_realistic_environments`: Daedalus and its eight hosts, the command-and-control server and the node-and-action pair, manipulated value, perceived state, the three deception actions, the scripted red agent, the three-day budget, and the three extensions in progress.

`td_sim2sim_before_sim2real`: the kill-chain schema, adversarial domain confusion, the goal-flag bypass, encoder pretraining from a random policy, and the source-and-target pairing across CyberWheel, NetSecGame, CyberBattleSim and NASim's emulation mode.

The check that closes this file is the two-way one. Every entry above names a nearest neighbour and each of those neighbours has to name it back. Sim2real names sim2sim and sim2sim names sim2real. Transfer names backend and backend names transfer. Realistic names fidelity and fidelity names realistic. Drift names the transfer gap and the transfer gap has to name drift. Where a distinction runs one way only, one of the two definitions is wrong, and the transfer pair is the one to watch, because it is the pair where the two senses never appear on the same page.

6 Next

Three questions carry into year two and none of them is answered by building more environment.

“A transferred policy that scores well has either learned the task or learned the benchmark, and the score alone will not say which.”

website, Next, winning for the right reasons

“Knowing when a deployed policy has aged out of usefulness, before it fails rather than after, is an open detection problem”

website, Next, drift after deployment

“Bounding worst-case behavior under model error, so that being wrong degrades the defender gracefully instead of catastrophically, is the last open piece”

website, Next, behavior while the model is wrong

The first is a measurement question and it needs an adversarial test rather than a better benchmark. The second is a detection question and the hard part is the qualifier: shift that matters to this policy, not shift in general. The third is a guarantee, and it is the only one of the three that has to hold at a moment when everything else in this file has already failed.

If you want the criteria rather than the vocabulary, Metrion is where they live.